Companion — Kestrel user guide

This module routes regions of interest to DeckLink outputs from a Stream Deck or any other Bitfocus Companion surface, through Kestrel. You can take any region to any output, cycle regions on an output, fire a salvo and kill every output at once, with live crosspoint tally.

The README covers installing the module. This guide is about building a page with it.

> **Before you rely on this:** the module's 21 automated checks run against a **fake Kestrel**, so they show the module's own shape is right but not that the two ends agree. Kestrel itself has been verified against real SDI hardware; this module driving it hasn't been used in a show. When either side changes, the live test against a Kestrel that's actually running is the one that matters.
>
> **If the connection comes up with no actions, variables or presets at all**, you're on a version that dies during startup. Upgrade to the newest release before you look at anything else.
>
> This module was built with AI assistance, directed and reviewed by a human author.

Connecting

Kestrel's control API: the address along the bottom of Kestrel's own window, `127.0.0.1:9720` by default.

> **There's no authentication.** Anyone who can reach that port can re-route every output and kill the lot at once. Keep it on a management interface, not on a show network anyone can join.

Max crosspoint presets in the connection config caps how many buttons a big show generates.

The crosspoint

Take region to output is what this is all about: one region of the input picture, onto one SDI output.

The dropdown has a **— clear —** entry, which drops the crosspoint **without stopping the output**, so it falls back to its idle fill. That matters on a rig where something downstream is watching for signal, because clearing a crosspoint isn't the same as killing an output.

Cycle the region on an output steps forward or back through the regions, and can include "nothing routed" in the cycle if you want. It's there for a surface with fewer buttons than the show has

regions.

Salvo puts one region on *every* output at once. **Outputs on / off** is the global kill.

Set an output's idle fill picks black, bars or the full input for when nothing's routed. Bars is the one to go for when somebody downstream needs to see that the path is alive.

Tally, and the two feedbacks that differ

Crosspoint is taken colours the button **in the region's own colour, taken from Kestrel**, so the surface matches the operator's screen and nobody has to keep a second colour scheme up to date.

It **dims rather than clears** when the outputs are killed. The routing's still what it was, it just isn't going anywhere, and that's worth knowing during a kill rather than staring at a blank page.

Output is carrying a region is the stricter one: it's only true when that output is *properly on air*. If it's muted, unrouted or the input's gone, it's false.

Input is locked deserves a button of its own. When it's false, **every routed output is carrying black**, and nothing else on the surface will tell you so.

Output is scaling too far warns you above a threshold you set. It's the failure that doesn't announce itself: everything's routed correctly and a region is being blown up further than it can take.

Two things that will not do what you expect

It's safe to hold down a nudge button. Kestrel clamps at the edge of the frame rather than refusing the command, so a held button walks the region to the edge and stops there.

Killing the transmitter doesn't remove the signal. A DeckLink output carries on sending locked black long after the process that opened it has exited. Measured twenty seconds after the fact, a looped-back input still reported a good, locked 1080p50 signal. (Nobody told the card.)

So **"outputs off" is no way to prove to a downstream device that the source has gone.** If that's what you're testing, pull the cable.

Building a surface that fails safe

1. **Input is locked**, big and obvious. It's the one failure that makes every other light on the page a lie.

2. **Crosspoint tally in Kestrel's own region colours**, so the surface and the screen agree.
3. **Output-is-carrying** on anything that reads as "on air", rather than the crosspoint tally.
4. **Scaling-too-far with a sensible threshold**, set from what the show's outputs can take.
5. **The global kill somewhere you have to mean to reach.** It's one press with no confirmation.

Troubleshooting

Symptom	Cause
No actions or presets at all	The module died during startup. Upgrade.
Everything is routed and every output is black	The input is not locked. Add that feedback.
Tally is dim rather than lit	Outputs are killed. The routing is intact.
A downstream device still sees signal after outputs off	The card keeps transmitting after the process exits. This is not the module.
A region will not move any further	It is clamped at the frame edge, by design.
The picture is soft on one output	Scaling too far — the feedback exists for exactly this.

See also

- README: installing, and the full list of actions, feedbacks and variables
- `companion/HELP.md`: the same material, in Companion's help panel
